ngmi

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Education

University of bumfuck no where — bumfuck, nowhere
Bachelor of Science, Computer Engineering (GPA: 3.98/4) — *May 2026*

Work Experience

Robinhood — May 2025 - Present
Software Engineer Intern — *Toronto, ON*

- i will update this shortly

Same research lab — May 2024 - Dec 2024
Undergraduate Research Assistant — *no where*

- Achieved 85% accuracy in predicting cell viability from 10,000+ diffraction images using a custom CNN, establishing a viable, non-invasive measurement method for cell health, (also published 2 papers and idk if i i should put it somewhere)
- Built an interactive model validation GUI in Python that rendered SHAP values to provide real-time explanations of prediction drivers, enabling researchers to debug model behavior on a per-cell basis
- Architected an end-to-end data processing pipeline that boosted morphological parameter extraction accuracy by 17% and cut pre-processing runtime in half (50% reduction) using Python's multiprocessing library
- Replaced a high-risk system of manually-shared hard drives with a centralized MySQL database, eliminating data loss and versioning conflicts, and enabling complex, cross-experiment queries for the first time

Research Lab — May 2023 - Aug 2023
Undergraduate Research Assistant — *no where*

- Engineered a general-purpose Python control framework for the lab's oscilloscope, abstracting low-level hardware commands into a high-level API that empowered researchers to script and automate entire experimental sequences.
- Slashed data analysis runtime by over 90% (1 hour to <6 minutes) by refactoring legacy Python scripts, replacing inefficient loops with vectorized NumPy operations
- Developed a novel classification system using a Gaussian Mixture Model to determine cell viability from time-series impedance data, creating a probabilistic tool for assessing sample health

Projects

AI App Builder (Next.js, TypeScript, Prisma, Docker)

- Developed an app to generate and deploy web apps from user prompts, integrating a usage-based Stripe billing system to monetize user compute time and API token consumption
- Implemented asynchronous background job processing for AI model inference, reducing response times and preventing UI blocking during long-running code generation tasks
- Utilized Docker for user code sandboxing, secure code execution and application previews

Collaborative Document Editor (React, TypeScript, Socket.io)

- Engineered a collaborative text editor for up to 20 simultaneous users by broadcasting text operations, cursor locations, and online statuses, via WebSockets, achieving a sub-250ms synchronization latency
- Integrated Gemini to provide real-time AI-powered suggestions for text, formatting and content generation

Distributed Image Processing (C++, OpenMPI, OpenCV)

- Accelerated the processing of gigapixel-scale images by 3.8x compared to a single-threaded approach by designing an OpenMPI pipeline that partitioned data across a 4-node cluster

Skills

- **Programming Languages**: Python, TypeScript, JavaScript, C/C++, HTML/CSS, Java
- **Frameworks**: React, Next.js, PyTorch, Flask, Django, Express.js
- **Cloud**: AWS, GCP, Docker